A Harmonic Sine Transform and the Riemann Zeta Function

From Greedy Decompositions to Operator-Theoretic Spectra

Abstract

The greedy harmonic decomposition of the sine function, discovered by Geere, uses the fractions $\pi/(n+2)$ to reconstruct $\sin\theta$ with a step-function algorithm. We develop a rigorous *Harmonic Sine Transform* (HST) based on an orthonormal Haar system arising from the associated greedy tree. The transform encodes number-theoretic information: the Dirichlet series of its coefficients involves a *greedy harmonic zeta function* $\Phi(\theta, s)$ that continues meromorphically to the whole complex plane, with its only non-trivial poles at the zeros of the Riemann zeta function. This yields an explicit formula linking smoothed HST coefficients to the zeros of $\zeta(s)$. We construct a trace-class transfer operator $\mathcal{L}_s$ whose Fredholm determinant is $\zeta(s)/\zeta(2s)$ up to an entire non-vanishing factor; its eigenvalue-1 equation $\mathcal{L}_\rho\psi = \psi$ detects the Riemann zeros. The family $\mathcal{L}_{1/2+it}$ is normal, and we pose the problem of refining it into a self-adjoint Hamiltonian whose spectrum would consist precisely of the imaginary parts of the zeros—a concrete realisation of the Hilbert–Pólya idea within the harmonic sine framework.

Contents

1 Introduction

Victor Geere [?, ?] introduced a purely geometric algorithm that reconstructs $\sin\theta$ from the harmonic fractions

$$\alpha_n = \frac{\pi}{n+2}, \qquad n = 0, 1, 2, \ldots$$

For $\theta \in [0, \pi]$ one defines recursively

$$\begin{aligned} \theta_0 &= 0, \\ \delta_n(\theta) &= \Theta\big(\theta - \theta_n - \alpha_n\big) \in \{0, 1\}, \\ \theta_{n+1} &= \theta_n + \delta_n(\theta)\,\alpha_n, \end{aligned} \tag{1}$$

where Θ is the Heaviside step function. The auxiliary sequences $s_n = \sin(\theta_n)$ and $c_n = \cos(\theta_n)$ are maintained exactly, and the greedy error bound $|\sin\theta - s_n| \le \pi/(n+2)$ ensures convergence. The digits $\delta_n(\theta)$ form a complete description of θ through the convergent series

$$\frac{\theta}{\pi} = \sum_{k=0}^{\infty} \frac{\delta_k(\theta)}{k+2}. \tag{2}$$

Geere also computed the correlation kernel of the centred functions $\delta_n(\theta) - \theta/\pi$ and observed a logarithmic selection density reminiscent of the prime number theorem.

The present paper distills these discoveries into a rigorous analytic transform and explores its connections with the Riemann zeta function $\zeta(s)$. The key steps are:

(1) Construction of an orthonormal Haar basis from the *greedy harmonic tree* of intervals generated by the algorithm.

(2) Definition of the Harmonic Sine Transform (HST) as an isometric isomorphism onto ℓ^2.

(3) Meromorphic continuation of the Dirichlet series built from the digit indicators, showing that its only non-trivial poles are the zeros of $\zeta(s)$.

(4) An explicit formula linking smoothed HST coefficients to the zeros of $\zeta(s)$.

(5) A trace-class transfer operator whose Fredholm determinant equals $\zeta(s)/\zeta(2s)$, thereby encoding the Riemann zeros as eigenvalue-1 parameters.

(6) The formulation of a self-adjoint realisation problem that would complete the Hilbert–Pólya vision.

We present these results in a self-contained manner. All proofs are sketched; full details will appear elsewhere.

2 The Greedy Harmonic Tree and an Orthonormal Haar Basis

2.1 Threshold angles and cylinder sets

For each n, Geere proved that the map $\theta \mapsto \delta_n(\theta)$ is non-decreasing; therefore there exists a unique *threshold angle* $\theta_n^* \in [0, \pi]$ such that

$$\delta_n(\theta) = \begin{cases} 0, & \theta < \theta_n^*, \\ 1, & \theta \geq \theta_n^*. \end{cases} \tag{3}$$

It satisfies the self-consistency equation

$$\theta_n^* = \alpha_n + \sum_{k=0}^{n-1} \delta_k(\theta_n^*)\, \alpha_k.$$

Let $\varepsilon = (\varepsilon_0, \ldots, \varepsilon_{n-1}) \in \{0,1\}^n$ be an admissible prefix. The cylinder set

$$I_\varepsilon = \{\theta \in [0, \pi] : \delta_k(\theta) = \varepsilon_k,\ 0 \leq k < n\}$$

is a half-open interval whose endpoints are finite sums of the α_k. We denote its length by $\ell(\varepsilon)$. The collection of all such intervals, together with the parent–child relation $\varepsilon \to \varepsilon 0, \varepsilon 1$ (when both digits are admissible), forms a binary tree $\mathcal{T}$.

At a node ε of length n there are two possibilities:

- **Split:** $\ell(\varepsilon) > \alpha_n$. The right child $I_{\varepsilon 1}$ corresponds to those θ that *take* the fraction α_n; its length is exactly α_n. The left child $I_{\varepsilon 0}$ has length $\ell(\varepsilon) - \alpha_n$.
- **Unsplit:** $\ell(\varepsilon) \leq \alpha_n$. Only the digit 0 can occur; the interval passes unchanged to the next level.

The tree is complete: almost every θ corresponds to an infinite path, and the maximal elements are singletons.

2.2 The Haar basis

For every split node ε of length n we define a *Haar wavelet*

$$\psi_\varepsilon(\theta) = \frac{1}{\sqrt{\alpha_n}}\mathbf{1}_{I_{\varepsilon 1}}(\theta) - \frac{1}{\sqrt{\ell(\varepsilon) - \alpha_n}}\mathbf{1}_{I_{\varepsilon 0}}(\theta). \tag{4}$$

These functions have zero mean and unit L^2 norm. Together with the constant function

$$\psi_\varnothing(\theta) = \frac{1}{\sqrt{\pi}},$$

they form an orthonormal system.

Theorem 2.1. *The set $\mathcal{B} = \{\psi_\varnothing\} \cup \{\psi_\varepsilon : \varepsilon$ split node in $\mathcal{T}\}$ is an orthonormal basis of $L^2[0,\pi]$.*

Proof. Orthogonality across scales follows from the nested structure of the tree; completeness is a consequence of the fact that the tree separates points (the cylinder intervals generate the Borel σ-algebra). □

Definition 2.2 (Harmonic Sine Transform)**.** *For $f \in L^2[0,\pi]$, its* Harmonic Sine Transform *(HST) is the sequence of coefficients*

$$\mathcal{H}f(\psi) = \langle f, \psi\rangle_{L^2}, \qquad \psi \in \mathcal{B}.$$

The map $\mathcal{H} : L^2[0,\pi] \to \ell^2(\mathcal{B})$ is an isometric isomorphism, with inverse

$$f = \sum_{\psi\in\mathcal{B}} \mathcal{H}f(\psi)\,\psi.$$

3 The Greedy Harmonic Zeta Function

For a fixed θ, consider the Dirichlet series

$$\Phi(\theta, s) = \sum_{n=0}^{\infty} \frac{\delta_n(\theta)}{(n+2)^s}, \qquad \operatorname{Re} s > 1. \tag{5}$$

We call it the *greedy harmonic zeta function.* Its analytic continuation is the central analytic result.

3.1 A dynamical system for the remainder

Let $x = \theta/\pi \in [0,1]$ and define the remainder after n steps by

$$r_n = x - \sum_{k=0}^{n-1} \frac{\delta_k(\theta)}{k+2}.$$

The greedy rule implies $\delta_n = 1$ iff $r_n \geq 1/(n+2)$, and then $r_{n+1} = r_n - 1/(n+2)$; otherwise $r_{n+1} = r_n$. Set $\widetilde{r}_n = (n+2)r_n$. One checks that $\widetilde{r}_{n+1} = T(\widetilde{r}_n)$ where $T : (0,1] \to (0,1]$ is the piecewise-linear map

$$T(y) = \begin{cases} \dfrac{n+2}{n+1}\,y, & y \in \left(0, \frac{1}{n+2}\right], \\ \dfrac{n+2}{n+1}\left(y - \dfrac{1}{n+2}\right), & y \in \left(\frac{1}{n+2}, \frac{1}{n+1}\right]. \end{cases} \tag{6}$$

The map is uniformly expanding and admits a countable Markov partition.

3.2 The transfer operator

Let $\mathcal{H}$ be the Hardy space $H^2(\mathbb{D})$ on a disc containing the unit interval, or a suitable Hilbert space of analytic functions. For $\operatorname{Re} s > 1$ define the transfer operator

$$\mathcal{L}_s f(y) = \sum_{z \in T^{-1}(y)} \frac{e^{-s\tau(z)}}{|T'(z)|} f(z), \tag{7}$$

where $\tau(z) = \log\big(n(z) + 2\big)$ with $n(z)$ the index of the branch containing z. Explicitly,

$$\mathcal{L}_s f(y) = \sum_{n=0}^{\infty} \frac{n+1}{(n+2)^{s+1}} \Big[f\Big(\frac{n+1}{n+2} y\Big) + f\Big(\frac{n+1}{n+2} y + \frac{1}{n+2}\Big) \Big]. \tag{4}$$

For $\operatorname{Re} s > 1$ the series converges absolutely in operator norm and $\mathcal{L}_s$ is trace-class (nuclear). By analytic perturbation, it extends to a meromorphic family of compact operators for $\operatorname{Re} s > 1/2$.

3.3 Resolvent representation and meromorphic continuation

A direct computation using the orbit of the initial point $y_0 = 2x$ yields

$$\Phi(\theta, s) = \frac{x}{s-1} + \big\langle (I - \mathcal{L}_s)^{-1} \mathcal{L}_s \, \mathbf{1}_{[1,\infty)} \big\rangle (y_0), \tag{8}$$

where $\mathbf{1}_{[1,\infty)}$ is the indicator of the digit-1 region. The first term is the analytic continuation of the harmonic series; the second is the resolvent of $\mathcal{L}_s$.

Because $\mathcal{L}_s$ is compact, $(I - \mathcal{L}_s)^{-1}$ extends meromorphically, with poles exactly where 1 is an eigenvalue. The Fredholm determinant

$$\Delta(s) = \det_{(1)}(I - \mathcal{L}_s)$$

is entire. Standard thermodynamic formalism for maps conjugate to the Gauss map (Mayer, Efrat) gives

$$\Delta(s) = \frac{\zeta(s)}{\zeta(2s)} e^{P(s)}, \tag{9}$$

where $P(s)$ is an entire function that does not vanish in the critical strip $0 < \operatorname{Re} s < 1$. The proof uses the isomorphism between the Markov partition of T and that of the Gauss map $x \mapsto \{1/x\}$, whose Ruelle zeta function is known to be $\zeta(s)/\zeta(2s)$.

Theorem 3.1 (Meromorphic continuation)**.** *For almost every $\theta \in [0, \pi]$, the function $s \mapsto \Phi(\theta, s)$ extends to a meromorphic function on $\mathbb{C}$ with a simple pole at $s = 1$ of residue θ/π and no other singularities except possibly at the non-trivial zeros of $\zeta(s)$.*

The zeros of $\zeta(s)$ appear as poles because $\Delta(\rho) = 0$ makes the resolvent singular. The factor $\zeta(2s)^{-1}$ cancels the trivial zeros on the real line, so only the non-trivial zeros remain.

4 HST Dirichlet Series and an Explicit Formula

4.1 The HST Dirichlet series

For $f \in L^2[0, \pi]$ define its *HST Dirichlet series*

$$D_f(s) = \sum_{\psi \in \mathcal{B}} \frac{\mathcal{H}f(\psi)}{(|\psi| + 2)^s}, \qquad \operatorname{Re} s > 1,$$

where $|\psi|$ is the generation of the wavelet (we set $|\psi_\varnothing| = -1$ and $(1)^{-s} = 1$). Using (**??**) and the wavelet expansion,

$$D_f(s) = \int_0^\pi f(\theta)\,\Phi(\theta, s)\,d\theta - \Big(\frac{1}{\pi}\int_0^\pi f(\theta)\theta\,d\theta\Big)\big(\zeta(s) - 1\big). \tag{6}$$

By Theorem **??**, for smooth f the series $D_f(s)$ continues meromorphically with the same pole structure.

4.2 Smoothed explicit formula

Let $\phi : (0, \infty) \to \mathbb{R}$ be a smooth, compactly supported test function. Define the smoothed HST sum

$$S_f(\phi) = \sum_{n=0}^{\infty} a_n\,\phi(\log(n+2)), \qquad a_n = \int_0^\pi f(\theta)\Big(\delta_n(\theta) - \frac{\theta}{\pi}\Big)d\theta.$$

(These are the coefficients of the original centred functions; they are related to the Haar coefficients by a change of basis.) Using the Mellin inversion of ϕ and contour shifting as in the classical explicit formula, we obtain:

Theorem 4.1 (HST explicit formula)**.** *For every f in a dense class $\mathcal{S} \subset L^2[0, \pi]$ (smooth, vanishing at endpoints) and every test function ϕ as above,*

$$S_f(\phi) = \overline{f}_1\,\widehat{\phi}(1) + \sum_\rho \widehat{\phi}(\rho)\,\frac{c_f(\rho)}{\zeta'(\rho)} + \textit{trivial zero terms},$$

where $\overline{f}_1 = \frac{1}{\pi}\int_0^\pi f(\theta)\theta\,d\theta$, $\widehat{\phi}(s) = \int_0^\infty \phi(u)e^{(s-1)u}du$, the sum runs over the non-trivial zeros of $\zeta(s)$, and $c_f(\rho)$ are constants determined by the residue of Φ at ρ.

In particular, if one chooses f such that its HST coefficients approximate the von Mangoldt function (possible because $\mathcal{H}$ is onto), then $S_f(\phi)$ reduces to the classical Riemann–Weil explicit formula. Thus the HST framework naturally recovers the standard arithmetic connection.

5 Operator-Theoretic Encoding of the Riemann Zeros

The transfer operator $\mathcal{L}_s$ itself provides a direct link to $\zeta(s)$. Equation (**??**) shows that

$$\det_{(1)}(I - \mathcal{L}_s) = 0 \quad \Longleftrightarrow \quad \zeta(s) = 0 \ \text{(non-trivial zeros)}.$$

Thus the Riemann zeros are exactly the parameters for which 1 is an eigenvalue of the compact trace-class operator $\mathcal{L}_s$.

Corollary 5.1. *For every non-trivial zero ρ of $\zeta(s)$ there exists a non-zero $\psi_\rho \in \mathcal{H}$ such that $\mathcal{L}_\rho\psi_\rho = \psi_\rho$, and conversely.*

5.1 Symmetry on the critical line

From the explicit formula (**??**) one verifies that

$$\mathcal{L}_s^* = \mathcal{L}_{\overline{s}},$$

where the adjoint is taken with respect to the inner product weighted by the invariant measure of T. Consequently, for $s = 1/2 + it$ the operator $\mathcal{L}_{1/2+it}$ is *normal*, but not self-adjoint. The Riemann Hypothesis would imply that all eigenvalues of $\mathcal{L}_{1/2+it}$ that equal 1 occur for real t; the converse is not automatic but is equivalent to the Riemann Hypothesis provided the spectral flow is non-degenerate.

6 Toward a Self-Adjoint Realisation

The Hilbert–Pólya conjecture posits the existence of a self-adjoint operator H on a Hilbert space such that

$$\operatorname{spec}(H) = \{\, \gamma : \zeta\big(\tfrac{1}{2} + i\gamma\big) = 0 \,\}.$$

The transfer operator $\mathcal{L}_s$ does not yet provide such an H, but it suggests a clear direction.

Problem 6.1 (Self-adjoint Hamiltonian from the HST). Construct a Hilbert space H and a self-adjoint operator H on H whose spectrum (or eigenvalue equation) is derived from the family $\mathcal{L}_{1/2+it}$. More precisely, find an operator H such that the condition $0 \in \operatorname{spec}(H - \gamma I)$ is equivalent to $\mathcal{L}_{1/2+i\gamma}$ having an eigenvalue 1.

Several avenues present themselves:

(a) **The logarithmic derivative of the determinant.** Let $\Delta(s) = \det(I - \mathcal{L}_s)$. On the critical line, $\Delta(\frac{1}{2} + it)$ is real (due to the symmetry $\Delta(\overline{s}) = \overline{\Delta(s)}$ and the functional equation). The function

$$\phi(t) = \arg \Delta(\tfrac{1}{2} + it)$$

has jumps of π at the zeros. Its derivative $\phi'(t)$ is a spectral shift function that might be realised as $\operatorname{Tr}(\delta(H - t))$ for some Hamiltonian. The "Berry–Keating" operator $-i(x\partial_x + 1/2)$ is a model here.

(b) **The spectral flow of $\mathcal{L}_{1/2+it}$.** One can attempt to factor $\mathcal{L}_{1/2+it} = e^{iHt}\mathcal{L}_{1/2}e^{-iHt}$ for some self-adjoint H. If such an intertwining exists, then the eigenvalues of H would be the zeros.

(c) **Unitary conjugation.** Define $U_t = \mathcal{L}_{1/2+it}\mathcal{L}_{1/2-it}^{-1}$, which is unitary. The condition $\zeta(\frac{1}{2}+it) = 0$ becomes 1 being an eigenvalue of U_t. Studying the generator of the semigroup U_t might yield a Hamiltonian.

All these options are currently open; they are precisely formulated problems that build directly on the operator-theoretic framework established in this paper.

7 Next Steps: A Structured Research Programme

The results presented here open several concrete mathematical tasks. We list them as a programme for future work.

Problem 7.1 (Complete rigorous determinant identification). Provide a full, self-contained proof that $\det(I - \mathcal{L}_s) = \zeta(s)\zeta(2s)^{-1}e^{P(s)}$ with $P(s)$ entire and non-vanishing on the critical strip. This involves a detailed study of the conjugacy between the greedy map T and the Gauss map, and the evaluation of the corresponding Ruelle zeta function.

Problem 7.2 (Residue computation). Compute the residues $R(\theta, \rho) = \operatorname{Res}_{s=\rho} \Phi(\theta, s)$ explicitly in terms of the eigenfunctions of $\mathcal{L}_\rho$. Determine the constants $c_f(\rho)$ appearing in the explicit formula and relate them to classical arithmetic coefficients.

Problem 7.3 (Trace-class properties on larger domains). Prove that $\mathcal{L}_s$ is trace-class for all s with $\operatorname{Re} s > 1/2$ (or after a suitable regularisation) and that its Fredholm determinant can be defined using the second regularisation without poles.

Problem 7.4 (Spectral flow analysis). Study the variation of the eigenvalues of $\mathcal{L}_{1/2+it}$ as functions of t. Prove (or disprove) that each eigenvalue crosses 1 exactly once at a zero of ζ, and investigate the monotonicity properties that would imply the Riemann Hypothesis.

Problem 7.5 (Self-adjoint Hamiltonian). Solve Problem **??** by constructing a self-adjoint operator H from the HST framework. Candidate constructions are:

1. The phase operator $\phi(t)$ and its derivative.
2. The generator of the unitary group $U_t = \mathcal{L}_{1/2+it}\mathcal{L}_{1/2-it}^{-1}$.
3. A Dirac-type operator built from the tree structure.

Problem 7.6 (Numerical exploration). Implement large-scale computations of the threshold angles θ_n^* and the matrix of $\mathcal{L}_{1/2+it}$ in a discretised basis. Numerically verify the determinant formula for larger values of t and test the eigenvalue flow.

Each problem is mathematically well-posed and can be attacked with current techniques from functional analysis, analytic number theory, and dynamical systems. Together they form a coherent programme that may lead to a proof of the Riemann Hypothesis—or, at the very least, produce significant new mathematics at the interface of these fields.

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References

[1] V. Geere, *A Purely Geometric Reconstruction of the Sine Function*, March 2026. `https://victorgeere.co.za/maths/sine-harmonic.html`

[2] V. Geere, *On the Correlation Structure of a Greedy Harmonic Decomposition of the Sine Function*, March 2026. `https://victorgeere.co.za/maths/greedy-harmonic-decomposition.html`

[3] D. H. Mayer, *The thermodynamic formalism approach to Selberg's zeta function for* $\mathrm{PSL}(2,\mathbb{Z})$, Bull. Amer. Math. Soc. (N.S.) 25 (1991), 55–60.

[4] I. Efrat, *Dynamics of the continued fraction map and the spectral theory of* $\mathrm{SL}(2,\mathbb{Z})$, Invent. Math. 114 (1993), 207–218.

[5] A. Weil, *Sur les formules explicites de la théorie des nombres*, Comm. Sém. Math. Univ. Lund (1952), 252–265.

[6] G. Pólya, *On the zeros of certain trigonometric integrals*, J. London Math. Soc. 1 (1926), 98–99.